
```

1. """Periodic Table of Elements, by Al Sweigart al@inventwithpython.com
2. Displays atomic information for all the elements.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: short, science"""
5.
6. # Data from https://en.wikipedia.org/wiki/List_of_chemical_elements
7. # Highlight the table, copy it, then paste it into a spreadsheet program
8. # like Excel or Google Sheets like in https://invpy.com/elements
9. # Then save this file as periodictable.csv.
10. # Or download this csv file from https://invpy.com/periodictable.csv
11.
12. import csv, sys, re
13.
14. # Read in all the data from periodictable.csv.
15. elementsFile = open('periodictable.csv', encoding='utf-8')
16. elementsCsvReader = csv.reader(elementsFile)
17. elements = list(elementsCsvReader)
18. elementsFile.close()
19.
20. ALL_COLUMNS = ['Atomic Number', 'Symbol', 'Element', 'Origin of name',
21.                'Group', 'Period', 'Atomic weight', 'Density',
22.                'Melting point', 'Boiling point',
23.                'Specific heat capacity', 'Electronegativity',
24.                'Abundance in earth\'s crust']
25.
26. # To justify the text, we need to find the longest string in ALL_COLUMNS.
27. LONGEST_COLUMN = 0
28. for key in ALL_COLUMNS:
29.     if len(key) > LONGEST_COLUMN:
30.         LONGEST_COLUMN = len(key)
31.
32. # Put all the elements data into a data structure:
33. ELEMENTS = {} # The data structure that stores all the element data.
34. for line in elements:
35.     element = {'Atomic Number': line[0],
36.                'Symbol': line[1],
37.                'Element': line[2],
38.                'Origin of name': line[3],
39.                'Group': line[4],
40.                'Period': line[5],
41.                'Atomic weight': line[6] + ' u', # atomic mass unit
42.                'Density': line[7] + ' g/cm^3', # grams/cubic cm
43.                'Melting point': line[8] + ' K', # kelvin
44.                'Boiling point': line[9] + ' K', # kelvin
45.                'Specific heat capacity': line[10] + ' J/(g*K)',
46.                'Electronegativity': line[11],
47.                'Abundance in earth\'s crust': line[12] + ' mg/kg'}
48.
49. # Some of the data has bracketed text from Wikipedia that we want to
50. # remove, such as the atomic weight of Boron:
51. # "10.81[III][IV][V][VI]" should be "10.81"
52.
53. for key, value in element.items():
54.     # Remove the [roman numeral] text:

```